# DNA Methylation Analysis Manual

## Supplementary Materials — Group 4

MSc in Bioinformatics — University of Bologna

June 2025

---

## Contents



---

# Preface

This document serves as an extensive technical guide for the DNA methylation analysis pipeline implemented by Group 4. It integrates theoretical foundations, detailed R code explanations, best practices, and troubleshooting tips to empower rigorous and reproducible Illumina HumanMethylation450K data analysis.

# 1 Glossary of Key Terms

### Glossary (Part 1)

**p-value** The probability of obtaining a result as extreme as the observed one under the null hypothesis.

**Detection p-value** Metric indicating if the probe signal is distinguishable from background noise (values ¿0.01 are often unreliable).

**Beta-value** Ratio of methylated to total signal, representing methylation level (range 0–1).

**M-value** $\text{Log}_2$ ratio of methylated to unmethylated signal; preferred for statistical testing due to improved variance properties.

**CpG site** DNA region where a cytosine nucleotide is followed by a guanine; key location for DNA methylation.

**Heatmap** A color-coded matrix visualization representing probe methylation across samples, often used for clustering analysis.

**Workspace** The R environment that stores all loaded objects during a session.

**Object** Any data structure stored in R, such as vectors, data frames, or lists.

**Function** A block of R code that performs a specific task.

**Variable** A name assigned to an object in R, enabling later reference.

**Script** A text file (.R) containing a sequence of R commands.

**Console** The interactive R prompt in RStudio or R.

**Data frame** A two-dimensional table-like structure with rows and columns.

**Vector** A one-dimensional sequence of data elements of the same type.

**List** A flexible container that can store multiple types of R objects.

## Glossary (Part 2)

**Loop** A control structure that repeats a block of code multiple times.

**Operator** A symbol (e.g., `+`, `-`, `>`, `<`, `==`) used to perform operations.

**False Discovery Rate (FDR)** Correction method that controls the expected proportion of false positives among significant results.

**Dye Bias** Technical artifact introduced by different dye efficiencies; corrected via normalization.

**Manifest file** Metadata file that maps probe IDs to genomic coordinates and annotations.

**RGChannelSet** minfi object storing raw red and green channel intensities.

**MethylSet** minfi object after background correction, ready for downstream analysis.

**Normalization** Process that adjusts data to remove technical variation while preserving biological differences.

**Batch effect** Unwanted variation introduced by experimental conditions (e.g., processing date, operator).

**Pipeline** An ordered sequence of analysis steps.

**Preprocessing** Steps like background correction and normalization applied to raw data to prepare it for analysis.

**Differential methylation** Statistical comparison of methylation levels between groups (e.g., disease vs. control).

**t-test** A statistical test comparing the means of two groups to identify significant differences.

**Volcano plot** A scatter plot showing significance ($-\log_{10}$ p-value) versus effect size (e.g., delta beta).

**Manhattan plot** Genome-wide plot showing $-\log_{10}$ p-values by chromosome position.

**PCA (Principal Component Analysis)** A technique to explore sample variance and detect batch effects.

**Pipe operator (%>%)** An operator from magrittr/dplyr to chain commands for cleaner syntax (not used in this workflow).

# 2 Theoretical Background

## DNA Methylation

DNA methylation is an epigenetic modification that occurs when a methyl group is added to the 5' position of cytosine residues within CpG dinucleotides. This modification is crucial for regulating gene expression and maintaining genome stability. Aberrant methylation is associated with diseases such as cancer and neurological disorders.

## Illumina 450K BeadChip

The Illumina HumanMethylation450K BeadChip enables high-throughput analysis of over 485,000 CpG sites across the genome. It uses bisulfite conversion to differentiate between methylated and unmethylated cytosines:

- **Infinium I**: Two separate probes (one for methylated, one for unmethylated) and single-color detection.
- **Infinium II**: A single probe with two-color detection to distinguish methylation status.

## Bisulfite Conversion

Bisulfite treatment converts unmethylated cytosines into uracils (read as thymines after PCR), while methylated cytosines remain unchanged. This differential conversion allows the array probes to detect methylation states.

| **Characteristic** | **Infinium I** | **Infinium II** |
|---|---|---|
| Number of Probes | Two (one for methylated, one for unmethylated) | One (single probe detects both) |
| Detection Channel | Single color per probe | Dual color within probe |
| CpG Coverage | Slightly lower | Slightly higher |
| Probe Design | Two separate bead types | One bead type |
| Signal Intensity | Higher dynamic range | Lower dynamic range |
| Normalization | Less dye bias, easier normalization | More dye bias, needs adjustment |
| Dye Bias Susceptibility | Lower | Higher |
| Interpretation | Raw signals easier to interpret | Requires correction for dye bias |

Table 1: Comparison of Infinium I and Infinium II chemistries on the Illumina 450K array.

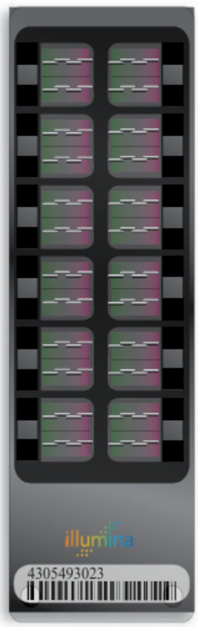


Figure 1: Example of an Illumina 450K bead chip.

# 3 R — A Deep Dive

## What is R?

R is a free, open-source programming language and environment specifically designed for statistical computing and data visualization. It is widely used in bioinformatics because of its:

- Flexibility in handling large and complex datasets typical of omics studies.
- Extensive ecosystem of packages dedicated to genomics and epigenetics (e.g. `minfi`, `limma`, `edgeR`).
- Integration with Bioconductor, providing standardized workflows for methylation analysis.
- Seamless interface with RStudio, an integrated development environment (IDE) that simplifies script writing, data visualization, and debugging.

R is particularly valued for its ability to chain together data preprocessing, statistical testing, visualization, and result export, making it a complete solution for DNA methylation analysis.

# The RStudio Interface

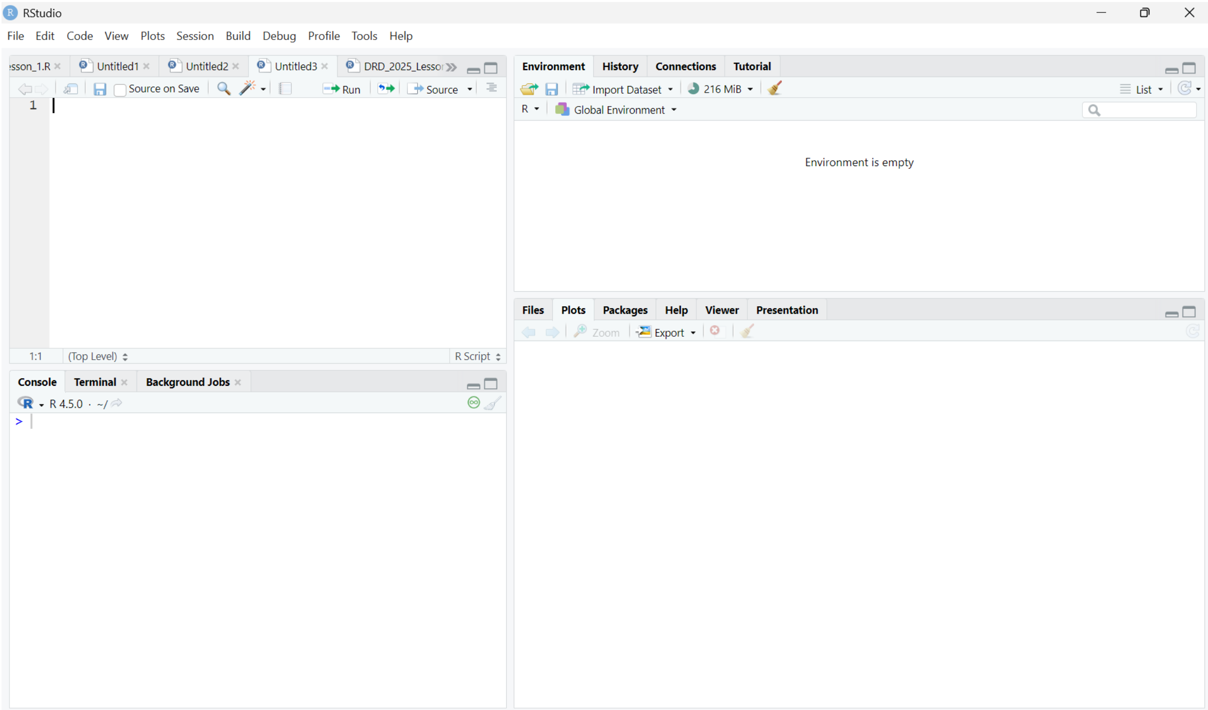


Figure 2: RStudio Interface — Console, Script Editor, Workspace, Plots.

## Tips and Useful Commands in R

To efficiently work with R, it is helpful to know some basic tips and commands:

- Use the assignment operator `<-` to create objects: `mydata <- read.csv("data.csv")`.
- Clean the workspace to avoid object conflicts: `rm(list = ls())`.
- Use the `#` symbol to comment and annotate code.
- Check installed packages with `installed.packages()` and load them with `library(pkgname)`.
- Explore data with commands like `str()`, `summary()`, `head()`, and `tail()`.
- Use pipes (%>%) from the `magrittr` or `dplyr` package to chain commands and improve code readability.
- For quick plotting, try `plot()`, while for more advanced graphics use `ggplot2`.

## Packages and Libraries

Packages extend R's functionality. Use:

Installing and Loading Packages

```
install.packages("ggplot2")
library(ggplot2)
if (!requireNamespace("BiocManager", quietly = TRUE))
    install.packages("BiocManager")
BiocManager::install("minfi")
```

### Data Structures in R

In R, data structures are fundamental for organizing and analyzing data. Here are the most common structures you will encounter in methylation analysis:

- **Vector**: A one-dimensional sequence of elements of the same type (numeric, character, logical).
- **Factor**: A special type of vector for categorical data with predefined levels.
- **Matrix**: A two-dimensional array of elements of the same type.
- **Data Frame**: A table-like structure with rows and columns, where each column can have a different type.
- **List**: A flexible container that can hold objects of different types and structures (e.g. data frames, vectors, even other lists).

Example:

```
# Vector
my_vector <- c(1, 2, 3, 4)

# Factor
my_factor <- factor(c("Control", "Treatment", "Control"))

# Data Frame
my_df <- data.frame(SampleID = c("A1", "A2"), Value = c(0.5, 0.7)
   )

# List
my_list <- list(Numeric = my_vector, Category = my_factor, Table
   = my_df)
```

Understanding these data structures is essential for efficient data manipulation and analysis in R.

## 4 About `minfi`

The `minfi` package is a comprehensive and essential R/Bioconductor toolkit for analyzing Illumina HumanMethylation arrays (450K and EPIC). It enables efficient processing and robust analysis of DNA methylation data by providing:

- Import of raw IDAT files using `read.metharray.exp()`, supporting flexible sample sheet integration.
- Quality control metrics, such as probe detection p-values via `detectionP()` and negative control plots, to ensure data reliability.
- Preprocessing methods, including background correction and functional normalization (`preprocessFunnorm()`), to correct technical biases and batch effects.
- Extraction of Beta-values and M-values using `getBeta()` and `getM()`, facilitating downstream statistical testing.

Overall, `minfi` supports a complete methylation analysis workflow, from raw data import to preprocessing and quality assessment, making it an essential package for reproducible and high-quality methylation studies.

**Note.** Additional R packages were also used to support data visualization and specialized analyses, including:

- `gplots` for heatmaps and hierarchical clustering.
- `qqman` for generating Manhattan plots.
- `factoextra` for PCA visualization.

# 5 Project Data

For this project, we analyzed DNA methylation data from the Illumina HumanMethylation450K array. Below is a summary of the key files and data objects used in the analysis:

| File/Data Object | Description |
|---|---|
| `SampleSheet.csv` | Sample metadata including ID, group (CTRL/DIS), sex, and batch. |
| `.idat files` | Raw fluorescence intensity data (red and green channels). |
| `Illumina450Manifest_clean.RData` | Cleaned manifest with probe IDs, genomic coordinates, and design type. |
| `RGChannelSet` | Raw data object created with `read.metharray.exp()`. |
| `MSet.raw` | Preprocessed object after background correction (`preprocessRaw()`). |
| `MSet.norm` | Normalized data using functional normalization (`preprocessFunnorm()`). |
| `Beta/M values` | Matrices of methylation quantification (`getBeta()`, `getM()`). |
| `DetectionP` | Detection p-values per probe/sample (`detectionP()`). |
| `final_ttest` | Data frame with raw p-values from group comparison. |
| `final_ttest_corr` | Data frame with corrected p-values (BH and Bonferroni). |
| `diff_meth_df` | Summary table of differentially methylated probes. |
| `pca_results` | PCA results object for sample clustering. |
| `Plots` | Figures including density plots, boxplots, PCA, volcano, Manhattan plots, and heatmaps. |

Table 2: Key files and data objects used in the Group 4 DNA methylation analysis.

Additionally, the following **assigned parameters** were used for the analysis:

| Parameter | Value |
|---|---|
| Group ID | 4 |
| Probe Address | 44666390 |
| Detection p-value cut-off | 0.01 |
| Normalization method | `preprocessFunnorm()` |
| Clustering methods | Average, Complete, Single linkage |

Table 3: Assigned parameters used in the analysis.

# 6 Pipeline Overview

An expanded step-by-step pipeline:

1. **Data Import**: Read raw .idat files using `read.metharray.exp()` to create an `RGChannelSet`.

2. **Annotation**: Use the cleaned manifest file (`Illumina450Manifest_clean.RData`) to map probe IDs and design types.

3. **Quality Control**: Calculate detection p-values (`detectionP()`) and generate quality control plots (`plotQC()`, `controlStripPlot()`).

4. **Preprocessing**: Apply background correction with `preprocessRaw()` and functional normalization with `preprocessFunnorm()`.

5. **Beta/M-value Calculation**: Calculate Beta-values (`getBeta()`) and M-values (`getM()`).

6. **Statistical Testing**: Perform t-tests (`t.test()`) comparing groups.

7. **Multiple Testing Correction**: Apply correction for multiple testing using `p.adjust()` (methods: BH, Bonferroni).

8. **Visualization**: Generate density plots, boxplots, PCA plots (`prcomp()`, `fviz_eig()`), volcano plots, Manhattan plots (`qqman`), and heatmaps (`heatmap.2()`).

# 7 The Analytical Pipeline — From Biological Question to Visualization

A successful DNA methylation study requires a well-designed workflow that systematically addresses every analytical step. Below, we outline a typical pipeline, emphasizing its logical flow from hypothesis formulation to data visualization:

1. **Biological Question**
   - Define the scientific goal, e.g., "Does methylation differ between disease and control groups?"
   - Establish hypotheses and experimental design (number of samples, replicates, conditions).

2. **Sample Collection and Preparation**
   - Extract DNA from relevant tissues or cell types.
   - Perform bisulfite conversion to differentiate methylated from unmethylated cytosines.

3. **Data Acquisition**
   - Use the Illumina HumanMethylation450K array (or EPIC) to profile methylation.
   - Generate raw data files (.idat) containing red and green channel intensities.

4. **Data Import into R**
   - Use `read.metharray.exp()` to read .idat files into an `RGChannelSet` object.
   - Inspect sample metadata (e.g., group, sex, batch) from `SampleSheet.csv`.

5. **Annotation**
   - Link probe IDs to genomic features using the cleaned manifest (`Illumina450Manifest_clean.RData`).
   - Exclude problematic probes (e.g., SNP-overlapping, cross-hybridizing).
6. **Quality Control**
   - Calculate detection p-values with `detectionP()` to filter unreliable probes.
   - Assess sample quality using `plotQC()` and `controlStripPlot()` for negative controls.
7. **Preprocessing**
   - Apply background correction using `preprocessRaw()`.
   - Normalize data using functional normalization with `preprocessFunnorm()` to mitigate technical biases and batch effects.
8. **Methylation Quantification**
   - Calculate Beta-values with `getBeta()` and M-values with `getM()`.
9. **Statistical Testing**
   - Perform t-tests for group comparisons (`t.test()`).
10. **Multiple Testing Correction**
    - Correct p-values for multiple testing using `p.adjust()` (methods: BH and Bonferroni).
11. **Visualization**
    - Generate density plots, boxplots, PCA plots (`prcomp()`, `fviz_eig()`), volcano plots, Manhattan plots (`qqman`), and heatmaps (`heatmap.2()`).
12. **Biological Interpretation**
    - Map significant CpGs to genes and pathways.
    - Integrate methylation findings with external datasets (e.g., gene expression, clinical data).

This pipeline illustrates how each step builds upon the previous one, transforming raw experimental data into biologically meaningful insights.

# 8 Function Reference

## Data Import

```
read.metharray.exp() | Imports raw .idat files into an RGChannelSet object.
```

### Annotation

```
Illumina450Manifest_clean | Cleaned manifest file for probe annotation.
```

### Quality Control

```
detectionP() | Computes detection p-values per probe.
plotQC() | Visualizes sample quality metrics.
controlStripPlot() | Visualizes negative control probes.
```

### Preprocessing

```
preprocessRaw() | Applies background correction.
preprocessFunnorm() | Performs functional normalization.
```

### Beta/M-value Calculation

```
getBeta() | Calculates Beta-values.
getM() | Calculates M-values.
```

### Statistical Testing

```
t.test() | Tests for group differences.
```

### Multiple Testing Correction

```
p.adjust() | Adjusts p-values for multiple testing (BH, Bonferroni).
```

### Visualization

```
heatmap.2() | Creates heatmaps with clustering.
fviz_eig() | Plots PCA eigenvalues.
manhattan() | Generates Manhattan plots.
```

# 9 Tips and Troubleshooting

- **Object Not Found**: Check spelling carefully (case-sensitive) and ensure that the object was created in the current R session. Use `ls()` to list all objects in the workspace.
- **Package Not Loaded**: Always load packages with `library()` before using their functions. For Bioconductor packages, remember to install via `BiocManager::install()` first.

- **Dimension Mismatch**: Check that data frames or matrices used in functions have the same number of rows/columns or match the expected structure. Use `dim()` and `head()` to inspect.
- **Warning Messages**: R issues warnings (e.g. `Warning: NAs introduced by coercion`) that do not stop code but may indicate potential problems. Review warnings carefully with `warnings()` and adjust code accordingly.
- **Missing Values (NAs)**: Check for missing values with `is.na()` and decide on appropriate handling (e.g. remove or impute).
- **Normalization Issues**: If downstream analyses seem off (e.g. unexpected sample clustering or dye bias), verify that background correction and normalization have been correctly applied. Re-check steps like `preprocessRaw()` and `preprocessFunnorm()`.
- **Dye Bias**: A technical artifact especially in Infinium II probes. Use appropriate normalization methods to correct (e.g. functional normalization).
- **Version Conflicts**: Make sure R and packages are updated to avoid deprecated functions. Check package version with `packageVersion("packagename")`.
- **Visualization Errors**: If plots look wrong or incomplete, check that the data is correctly structured and complete. Use `ggplot2` syntax carefully.
- **Unexpected Results**: Always verify that the correct samples and group labels were assigned during preprocessing. Use `table()` to cross-check factors like group and batch.

# 10 Conclusion

This manual is designed to guide users through the process of DNA methylation data analysis, from data import to visualization and interpretation. Our project demonstrates the critical role of bioinformatics in modern molecular biology and highlights the power of reproducible research. By using robust tools such as R and the minfi package, we can process large datasets efficiently and gain valuable insights into epigenetic regulation and disease mechanisms.

The expected results include the identification of differentially methylated regions (DMRs) that may serve as biomarkers for disease, as well as the discovery of new epigenetic patterns associated with clinical outcomes. This work contributes to the broader goal of personalized medicine, where DNA methylation signatures can inform both diagnostics and therapeutic strategies.

We hope this document empowers other researchers to replicate our pipeline and adapt it to their own datasets, fostering collaboration and accelerating discoveries in the field of epigenetics.